Best Management Practice Efficiency References for the Pollutant Load Estimation Tool

Updated August 2023

This document includes the following tables to document references used to determine best management practice (BMP) efficiency values for the pollutant load estimation tool (PLET). The last section of the document compiles all references used across land use types in alphabetical order.

Table 1 Cropland and Pastureland BMP Efficiencies and References

Table 2 Urban BMP Efficiencies and References

Table 3 Feedlots and Forest BMP Efficiencies and References

Table 1 Cropland and Pastureland BMP Efficiencies and References

BMP Name	Landuse Type	Pollutant	Mean	Min	Max	References
Bioreactor	Cropland	TN	0.45	0.13	0.72	Christianson et al. 2016, Husk et al. 2017, Woli et al. 2010, DeBoe et al. 2017, Tetra Tech 2016
Buffer - Forest (100ft wide)	Cropland	TN	0.49	0.19	0.92	Udawatta et al. 2018, Jontos 2004, Lee et al. 2003, USEPA 2010
Buffer - Forest (100ft wide)	Cropland	TP	**0.47**	**0.080**	**0.93**	Merriman et al. 2009, Udawatta et al. 2018, Jontos 2004, Lee et al. 2003, USEPA 2010, Jontos 2004
Buffer - Forest (100ft wide)	Cropland	Sediment	**0.60**	**0.19**	**0.95**	Udawatta et al. 2018, Schmitt et al. 1999, Yuan et al. 2009, Lee et al. 2003, USEPA 2010
Buffer - Grass (35ft wide)	Cropland	TN	0.34	0.13	0.46	USEPA 2010
Buffer - Grass (35ft wide)	Cropland	TP	0.44	0.30	0.47	USEPA 2010
Buffer - Grass (35ft wide)	Cropland	Sediment	0.53	0.40	0.60	USEPA 2010
Conservation Tillage 1 (30-59% Residue)	Cropland	TN	**0.070**	**0.040**	**0.10**	CBP 2016a
Conservation Tillage 1 (30-59% Residue)	Cropland	TP	0.36	0.020	0.60	CBP 2016a
Conservation Tillage 1 (30-59% Residue)	Cropland	Sediment	**0.46**	**0.41**	**0.50**	Christianson et al. 2016, CBP 2016a
Conservation Tillage 2 (equal or more than 60% Residue)	Cropland	TN	**0.13**	**0.12**	**0.14**	CBP 2016a
Conservation Tillage 2 (equal or more than 60% Residue)	Cropland	TP	0.69	0.11	0.90	Christianson et al. 2016, CBP 2016a
Conservation Tillage 2 (equal or more than 60% Residue)	Cropland	Sediment	**0.79**	**0.79**	**0.79**	CBP 2016a
Contour Farming	Cropland	TN	**0.34**	**0.23**	**0.49**	Evans, Lehning and Corradini 2003, USEPA 1993
Contour Farming	Cropland	TP	**0.46**	**0.40**	**0.55**	Merriman et al. 2009, Evans, Lehning and Corradini 2003, USEPA 1993
Contour Farming	Cropland	Sediment	**0.41**	**0.40**	**0.41**	Evans, Lehning and Corradini 2003, USEPA 1993
Controlled Drainage	Cropland	TN	0.39	0.33	0.45	Christianson et al. 2016, Evans et al. 1996
Controlled Drainage	Cropland	TP	0.35	0.35	0.35	Evans et al. 1996
Cover Crop 1 (Group A Commodity) (High Till only for Sediment)	Cropland	TN	0.0078	0	0.013	CBP 2015, CBP 2017
Cover Crop 2 (Group A Traditional Normal Planting Time) (High Till only for TP and Sediment)	Cropland	TN	0.20	0.20	0.20	CBP 2015, CBP 2017
Cover Crop 2 (Group A Traditional Normal Planting Time) (High Till only for TP and Sediment)	Cropland	TP	0.070	0.070	0.070	CBP 2015, CBP 2017
Cover Crop 2 (Group A Traditional Normal Planting Time) (High Till only for TP and Sediment)	Cropland	Sediment	0.10	0.10	0.10	CBP 2015, CBP 2017
Cover Crop 3 (Group A Traditional Early Planting Time) (High Till only for TP and Sediment)	Cropland	TN	0.20	0.20	0.20	CBP 2015, CBP 2017
Cover Crop 3 (Group A Traditional Early Planting Time) (High Till only for TP and Sediment)	Cropland	TP	0.15	0.15	0.15	CBP 2015, CBP 2017

Table 1 Cropland and Pastureland BMP Efficiencies and References

BMP Name	Landuse Type	Pollutant	Mean	Min	Max	References
Cover Crop 3 (Group A Traditional Early Planting Time) (High Till only for TP and Sediment)	Cropland	Sediment	0.20	0.20	0.20	CBP 2015, CBP 2017
Land Retirement	Cropland	TN	0.90	0.83	0.96	Evans, Lehning and Corradini 2003, Christianson et al. 2016
Land Retirement	Cropland	TP	0.81	0.56	0.98	Evans, Lehning and Corradini 2003, Christianson et al. 2016
Land Retirement	Cropland	Sediment	0.95	0.92	0.98	Evans, Lehning and Corradini 2003
Nutrient Management 1 (Determined Rate)	Cropland	TN	0.15	0.15	0.15	CBP 2016b
Nutrient Management 1 (Determined Rate)	Cropland	TP	0.45	0.45	0.45	Christianson et al. 2016, USEPA 1993, USEPA 2010
Nutrient Management 2 (Determined Rate Plus Additional Considerations)	Cropland	TN	**0.22**	**0.090**	**0.70**	Christianson et al. 2016, Lam et al. 2011, Shepard 2005, Evans, Lehning and Corradini 2003, USEPA 1993
Nutrient Management 2 (Determined Rate Plus Additional Considerations)	Cropland	TP	0.56	0.56	0.56	Determined based on input from Tetra Tech and CBP 2016b
Streambank Stabilization and Fencing	Cropland	TN	0.75	0.75	0.75	USEPA 1993
Streambank Stabilization and Fencing	Cropland	TP	0.75	0.75	0.75	USEPA 1993
Streambank Stabilization and Fencing	Cropland	Sediment	0.75	0.75	0.75	USEPA 1993
Terrace	Cropland	TN	**0.27**	**-6.0E-03**	**0.67**	Tuppad et al. 2010, Gassman et al. 2006, Evans, Lehning and Corradini 2003, USEPA 1993
Terrace	Cropland	TP	**0.31**	**0.060**	**0.70**	Tuppad et al. 2010, Gassman et al. 2006, Makarewicz et al. 2015, Evans, Lehning and Corradini 2003, USEPA 1993
Terrace	Cropland	Sediment	**0.41**	**-0.037**	**0.92**	Yang, et al. 2012, Tuppad et al. 2010, Gassman et al. 2006, Makarewicz et al. 2015, Evans, Lehning and Corradini 2003, USEPA 1993
Two-Stage Ditch	Cropland	TN	0.12	0.12	0.12	Tetra Tech 2016
Two-Stage Ditch	Cropland	TP	0.28	0.28	0.28	Tetra Tech 2016
30m Buffer with Optimal Grazing	Pastureland	TN	0.16	0.46	0.36	Chaubey et al. 2010
30m Buffer with Optimal Grazing	Pastureland	TP	0.65	0.66	0.65	Chaubey et al. 2010
Alternative Water Supply	Pastureland	TN	**0.18**	**0.050**	**0.30**	USEPA 2010, Oklahoma NPS Coordinator 2014
Alternative Water Supply	Pastureland	TP	**0.13**	**0.00**	**0.30**	USEPA 2010, Oklahoma NPS Coordinator 2014, Sheshukov et al. 2016
Alternative Water Supply	Pastureland	Sediment	**0.20**	**0**	**0.54**	USEPA 2010, Oklahoma NPS Coordinator 2014, Sheshukov et al. 2016, Line et al. 2000, Sheffield et al. 1997
Critical Area Planting	Pastureland	TN	0.18	0.050	0.30	USEPA 2010, Oklahoma NPS Coordinator 2014
Critical Area Planting	Pastureland	TP	0.20	0.10	0.30	USEPA 2010, Oklahoma NPS Coordinator 2014
Critical Area Planting	Pastureland	Sediment	0.42	0.14	0.70	USEPA 2010, Oklahoma NPS Coordinator 2014
Forest Buffer (minimum 35 feet wide)	Pastureland	TN	0.45	0.19	0.65	USEPA 2010
Forest Buffer (minimum 35 feet wide)	Pastureland	TP	0.40	0.30	0.45	USEPA 2010
Forest Buffer (minimum 35 feet wide)	Pastureland	Sediment	0.53	0.40	0.60	USEPA 2010
Grass Buffer (minimum 35 feet wide)	Pastureland	TN	0.87	0.78	0.95	Heathwaite et al. 1998, Lim et al. 2000
Grass Buffer (minimum 35 feet wide)	Pastureland	TP	**0.89**	**0.83**	**0.94**	Heathwaite et al. 1998, Lim et al. 2000
Grass Buffer (minimum 35 feet wide)	Pastureland	Sediment	0.65	0.17	0.98	Carline and Walsh 2007, Heathwaite et al. 1998, Lim et al. 2000
Grazing Land Management (rotational grazing with fenced areas)	Pastureland	TN	0.43	0.43	0.43	Evans, Lehning and Corradini 2003
Grazing Land Management (rotational grazing with fenced areas)	Pastureland	TP	0.26	0	0.59	ISU 2014, McDowell and Houlbrooke 2009, Owens, Edwards and Van Keuren, 1989

Table 1 Cropland and Pastureland BMP Efficiencies and References

BMP Name	Landuse Type	Pollutant	Mean	Min	Max	References
Heavy Use Area Protection	Pastureland	TN	0.18	0.05	0.30	USEPA 2010, Oklahoma NPS Coordinator 2014
Heavy Use Area Protection	Pastureland	TP	0.19	0.080	0.30	USEPA 2010, Oklahoma NPS Coordinator 2014
Heavy Use Area Protection	Pastureland	Sediment	0.33	0.10	0.50	USEPA 2010, Oklahoma NPS Coordinator 2014
Litter Storage and Management	Pastureland	TN	0.14	0.14	0.14	Oklahoma NPS Coordinator 2014
Litter Storage and Management	Pastureland	TP	0.14	0.14	0.14	Oklahoma NPS Coordinator 2014
Litter Storage and Management	Pastureland	Sediment	0.00	0.00	0.00	Oklahoma NPS Coordinator 2014
Livestock Exclusion Fencing	Pastureland	TN	**0.20**	**0.20**	**0.20**	Oklahoma NPS Coordinator 2014
Livestock Exclusion Fencing	Pastureland	TP	**0.43**	**0.02**	**0.76**	Carline and Walsh 2007, Line et al. 2000, Oklahoma NPS Coordinator 2014
Livestock Exclusion Fencing	Pastureland	Sediment	**0.64**	**0.20**	**0.87**	Oklahoma NPS Coordinator 2014, Carline and Walsh 2007, Line et al. 2000, Owens et al. 1996
Multiple Practices	Pastureland	TN	0.25	-0.39	0.89	Chaubey et al. 2010, Owens and Bonta 2004
Multiple Practices	Pastureland	TP	0.20	-5.2	0.66	Chaubey et al. 2010, Meals 2004, Sistani et al. 2008
Multiple Practices	Pastureland	Sediment	0.22	-1.0	0.61	Carline and Walsh 2007, Meals 2004, Sistani et al. 2008
Pasture and Hayland Planting (also called Forage Planting)	Pastureland	TN	0.18	-0.33	0.68	Owens et al. 1994, Owens, Edwards and Van Keuren, 1989, Oklahoma NPS Coordinator 2014
Pasture and Hayland Planting (also called Forage Planting)	Pastureland	TP	0.15	0.00	0.30	Oklahoma NPS Coordinator 2014, Owens, Edwards and Van Keuren, 1989
Prescribed Grazing	Pastureland	TN	0.41	0.090	0.92	Butler et al. 2007, USEPA 2010
Prescribed Grazing	Pastureland	TP	0.23	0.20	0.24	USEPA 2010
Prescribed Grazing	Pastureland	Sediment	0.33	0.30	0.40	USEPA 2010
Streambank Protection w/o Fencing	Pastureland	TN	0.15	0.15	0.15	USEPA 2010
Streambank Protection w/o Fencing	Pastureland	TP	0.22	0.22	0.22	USEPA 2010
Streambank Protection w/o Fencing	Pastureland	Sediment	0.58	0.30	0.85	USEPA 2010, Meals 2004
Streambank Stabilization and Fencing	Pastureland	TN	0.75	0.75	0.75	USEPA 1993
Streambank Stabilization and Fencing	Pastureland	TP	0.75	0.75	0.75	USEPA 1993
Streambank Stabilization and Fencing	Pastureland	Sediment	0.75	0.75	0.75	USEPA 1993
Use Exclusion	Pastureland	TN	**0.43**	**0.050**	**0.80**	Butler et al. 2008, USEPA 2010
Use Exclusion	Pastureland	TP	**0.08**	**0.080**	**0.080**	USEPA 2010
Use Exclusion	Pastureland	Sediment	**0.51**	**0.10**	**0.97**	Butler et al. 2008, USEPA 2010
Winter Feeding Facility	Pastureland	TN	0.35	0.35	0.35	Oklahoma NPS Coordinator 2014
Winter Feeding Facility	Pastureland	TP	0.40	0.40	0.40	Oklahoma NPS Coordinator 2014
Winter Feeding Facility	Pastureland	Sediment	0.40	0.40	0.40	Oklahoma NPS Coordinator 2014

Notes:

bolded values are updated as of 2023

Table 2 Urban BMP Efficiencies and References

BMP Name	Landuse Type	TN	TP	BOD	Sediment	Reference
Alum Treatment	Urban	0.60	0.90	0.60	0.95	U.S. EPA 2021
Bioretention facility	Urban	0.63	0.80	U	U	Davis et al 2001
Concrete Grid Pavement	Urban	0.90	0.90	U	0.9	MDEQ 1999
Dry Detention	Urban	0.30	0.26	0.27	0.58	MDEQ 1999
Extended Wet Detention	Urban	0.55	0.69	0.72	0.86	MDEQ 1999
Filter Strip-Agricultural	Urban	0.53	0.61	U	0.65	MDEQ 1999
Grass Swales	Urban	0.10	0.25	0.30	0.65	MDEQ 1999
Infiltration Basin	Urban	0.60	0.65	U	0.75	MDEQ 1999
Infiltration Devices	Urban	U	0.83	0.83	0.94	MDEQ 1999
Infiltration Trench	Urban	0.55	0.60	U	0.75	MDEQ 1999
LID*/Cistern	Urban	U	U	U	U	MDDER 1999
LID*/Cistern+Rain Barrel	Urban	U	U	U	U	MDDER 1999
LID*/Rain Barrel	Urban	U	U	U	U	MDDER 1999
LID/Bioretention	Urban	0.43	0.81	U	U	MDDER 1999
LID/Dry Well	Urban	0.50	0.50	0.70	0.90	MDDER 1999
LID/Filter/Buffer Strip	Urban	0.30	0.30	0.40	0.60	MDDER 1999
LID/Infiltration Swale	Urban	0.50	0.65	U	0.90	MDDER 1999
LID/Infiltration Trench	Urban	0.50	0.50	0.70	0.90	MDDER 1999
LID/Vegetated Swale	Urban	0.08	0.18	U	0.48	MDDER 1999
LID/Wet Swale	Urban	0.40	0.20	U	0.80	MDDER 1999
Oil/Grit Separator	Urban	0.05	0.05	U	0.15	MDEQ 1999
Porous Pavement	Urban	0.85	0.65	U	0.90	MDEQ 1999
Sand Filter/Infiltration Basin	Urban	0.35	0.50	U	0.80	MDEQ 1999
Sand Filters	Urban	U	0.38	0.40	0.83	MDEQ 1999
Settling Basin	Urban	U	0.52	0.56	0.82	MDEQ 1999
Vegetated Filter Strips	Urban	0.4	0.45	0.51	0.73	MDEQ 1999
WQ Inlet w/Sand Filter	Urban	0.35	U	U	0.80	MDEQ 1999
WQ Inlets	Urban	0.20	0.09	0.13	0.37	MDEQ 1999
Weekly Street Sweeping	Urban	U	0.06	0.06	0.16	MDEQ 1999
Wet Pond	Urban	0.35	0.45	U	0.60	MDEQ 1999
Wetland Detention	Urban	0.20	0.44	0.63	0.78	MDEQ 1999

Notes:

U no value

Table 3 Feedlots and Forest BMP Efficiencies and References

BMP Name	Landuse	TN	TP	BOD	Sediment	Reference
Diversion	Feedlots	0.45	0.70	U	U	U.S. EPA. 1993
Filter strip	Feedlots		0.85	U	U	U.S. EPA. 1993
Runoff Mgmt System	Feedlots		0.83	U	U	U.S. EPA. 1993
Solids Separation Basin	Feedlots	0.35	0.31	U	U	U.S. EPA. 1993
Solids Separation Basin w/Infilt Bed	Feedlots		0.80	0.85	U	U.S. EPA. 1993
Terrace	Feedlots	0.55	0.85	U	U	U.S. EPA. 1993
Waste Mgmt System	Feedlots	0.80	0.90	U	U	U.S. EPA. 1993
Waste Storage Facility	Feedlots	0.65	0.60	U	U	U.S. EPA. 1993
Road dry seeding	Forest	U	U	U	0.41	U.S. EPA. 1993
Road grass and legume seeding	Forest	U	U	U	0.71	U.S. EPA. 1993
Road hydro mulch	Forest	U	U	U	0.41	U.S. EPA. 1993
Road straw mulch	Forest	U	U	U	0.41	U.S. EPA. 1993
Road tree planting	Forest	U	U	U	0.50	U.S. EPA. 1993
Site preparation/hydro mulch/seed/fertilizer	Forest	U	U	U	0.71	Seyedbagheri, A. 1996
Site preparation/hydro mulch/seed/fertilizer/transplants	Forest	U	U	U	0.69	Seyedbagheri, A. 1996
Site preparation/steep slope seeder/transplant	Forest	U	U	U	0.81	Seyedbagheri, A. 1996
Site preparation/straw/crimp seed/fertilizer/transplant	Forest	U	U	U	0.95	Seyedbagheri, A. 1996
Site preparation/straw/crimp/net	Forest	U	U	U	0.93	Seyedbagheri, A. 1996
Site preparation/straw/net/seed/ fertilizer/transplant	Forest	U	U	U	0.83	Seyedbagheri, A. 1996
Site preparation/straw/polymer/ seed /fertilizer/transplant	Forest	U	U	U	0.86	Seyedbagheri, A. 1996

Notes:

U No Value

BMP Efficiencies Bibliography

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